

NORTHERN AROOSTOOK, ME, USA

WMO: 726083

Lat: 47.286N

Lon: 68.313W

Elev: 301

StdP: 97.76

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 04836

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.0	-22.6	-29.2	0.3	-24.2	-27.2	0.3	-21.8	12.9	-8.8	11.9	-11.5	4.5	320	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	28.0	19.5	26.3	18.4	24.5	17.7	21.1	25.5	20.1	24.2	19.2	22.9	4.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	14.8	23.0	18.7	14.0	22.3	17.7	13.2	20.8	62.6	25.4	59.1	24.2	55.8	22.8	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.7	8.6	DB	-28.1	30.9	2.8	1.7	-30.1	32.1	-31.7	33.0	-33.2	34.0	-35.2	35.2	(4)
(5)			WB	-28.2	22.9	2.6	1.0	-30.1	23.6	-31.6	24.2	-33.0	24.8	-34.9	25.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.0	-12.0	-10.2	-4.8	2.6	10.4	15.2	18.3	17.4	12.9	6.1	-0.8	-7.8	(6)	
	DBStd	11.58	6.59	6.08	5.77	4.30	4.32	3.77	3.02	3.17	4.11	4.24	4.93	5.78	(7)	
	HDD10.0	3011	683	565	459	225	48	3	0	0	16	135	324	552	(8)	
	HDD18.3	5313	941	798	717	473	248	109	37	56	170	378	574	811	(9)	
	CDD10.0	828	0	0	1	2	60	159	258	230	103	15	1	0	(10)	
	CDD18.3	88	0	0	0	0	2	14	37	28	7	0	0	0	(11)	
	CDH23.3	642	0	0	0	1	46	121	240	184	48	2	0	0	(12)	
	CDH26.7	126	0	0	0	0	8	24	48	37	8	0	0	0	(13)	
(14) Wind	WSAvg	4.2	4.6	4.6	4.8	4.6	4.2	3.7	3.3	3.3	3.7	4.2	4.4	4.5	(14)	
(15) Precipitation	PrecAvg	967	55	56	60	66	77	106	108	94	92	89	81	81	(15)	
	PrecMax	1245	118	110	113	131	136	167	151	190	193	170	170	138	(16)	
	PrecMin	736	25	22	17	28	39	61	33	30	27	40	12	45	(17)	
	PrecStd	136	24	20	25	29	26	30	31	43	40	35	35	29	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	5.7	12.0	19.2	27.6	29.2	30.6	29.9	27.7	21.4	15.0	7.4	(19)	
		MCWB	5.5	4.2	8.0	10.2	15.8	19.5	20.5	20.7	20.2	14.9	13.3	6.5	(20)	
	2%	DB	2.9	2.5	7.3	14.8	23.9	26.6	27.8	27.5	24.2	17.3	11.1	3.7	(21)	
		MCWB	2.0	0.9	2.8	7.8	14.2	18.1	19.9	19.4	18.4	12.7	9.3	2.8	(22)	
	5%	DB	0.4	0.9	4.8	12.0	21.2	24.2	26.0	25.3	22.2	14.9	8.2	1.8	(23)	
		MCWB	-0.5	-0.3	1.6	6.3	12.9	17.0	18.9	18.5	17.2	11.7	6.5	1.0	(24)	
	10%	DB	-2.3	-1.1	2.9	9.1	18.1	22.3	24.1	23.4	19.9	12.7	6.2	0.1	(25)	
		MCWB	-3.2	-2.2	0.6	4.9	11.4	15.9	18.2	17.5	15.8	9.9	4.6	-0.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	4.5	8.7	12.6	18.6	21.7	23.0	21.9	21.2	16.1	13.7	6.7	(27)	
		MCDB	6.0	5.3	10.8	17.0	24.8	27.4	26.8	27.2	26.2	18.7	15.0	7.2	(28)	
	2%	WB	2.2	1.4	3.9	9.2	16.0	19.8	21.3	20.7	19.5	14.0	9.6	3.0	(29)	
		MCDB	2.7	2.2	5.6	12.7	20.8	24.1	25.2	24.9	22.9	16.4	10.6	3.4	(30)	
	5%	WB	-0.5	-0.3	2.3	7.2	14.1	18.5	20.2	19.7	18.0	12.1	6.9	1.0	(31)	
		MCDB	0.3	0.5	4.3	10.8	18.9	22.2	24.2	23.6	20.8	14.3	7.9	1.5	(32)	
	10%	WB	-3.1	-2.2	0.7	5.4	12.5	17.2	19.2	18.8	16.5	10.4	4.7	-0.5	(33)	
		MCDB	-2.3	-1.3	2.8	9.0	17.4	20.3	23.1	22.1	19.2	12.4	6.3	0.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	8.5	8.6	8.4	10.3	10.1	9.6	9.9	9.8	7.8	6.2	6.7	(35)	
		MCDBR	10.4	8.4	9.9	13.4	15.3	13.2	11.6	12.0	12.0	11.4	9.1	8.1	(36)	
		MCWBR	10.4	7.4	7.2	8.5	7.8	6.8	5.6	5.7	6.9	7.7	8.0	8.0	(37)	
	5% WB	MCDBR	10.4	8.2	8.7	11.5	12.5	10.9	9.8	10.2	10.6	10.0	8.6	7.8	(38)	
		MCWBR	10.5	7.8	7.2	8.4	7.6	6.5	5.7	5.6	7.1	7.8	8.2	7.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.250	0.263	0.297	0.341	0.345	0.365	0.395	0.373	0.350	0.316	0.286	0.262	(40)	
	taud		2.344	2.299	2.271	2.360	2.439	2.381	2.314	2.379	2.441	2.532	2.528	2.395	(41)	
	Ebn at Noon		852	913	926	912	919	898	863	870	861	845	809	791	(42)	
	Edh at Noon		70	92	112	114	109	117	123	111	95	74	60	60	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.36	2.26	3.47	4.46	5.03	5.45	5.43	4.75	3.71	2.19	1.22	0.93	(44)	
	RadStd		0.17	0.27	0.25	0.37	0.52	0.36	0.51	0.31	0.27	0.20	0.14	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page